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Hybrid fixed angle rotor unmanned aerial vehicle with vertical takeoff and landing capabilities

Abstract

The disclosure relates to hybrid unmanned aerial vehicle (UAV) having vertical take-off and landing (VTOL) capabilities. Specifically, the disclosure relates to a stable hybrid fixed angle rotor arrays UAV having VTOL capabilities with hovering capabilities using rotors and cruising using fixed wing.

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Background/Summary

BACKGROUND

- (1) The disclosure is directed to a hybrid unmanned aerial vehicle (UAV) having vertical take-off and landing (VTOL) capabilities. Specifically, the disclosure is directed to a hybrid fixed and rotating wings UAV having VTOL capabilities.
- (2) Currently, UAVs are developed to complete a wide range of specialized tasks, such as, for example, combat, surveillance, delivery, search and rescue operations, industrial surveying and inspection, construction, mining, stockpiling, photogrammetry, aerial photography, cinematography, and video, live streaming, newsgathering, multispectral analysis, for vegetation, biological plant protection, asset perimeter inspection, transmission lines and pipelines inspection, interception of other UAVs, geodesy and cartography and other.
- (3) Typically, these UAVs were developed as multipurpose carrying platforms, to carry variable freight and/or payload. But any special application of UAV depended on special functional conditions and requirements, which in turn, are determined by the UAV's principal design solutions (design for X, or DFX). Among UAVs' determining DFX requirements are factors such as, flight duration, cruising altitude, payload type and weight, hovering ability and the like.
- (4) However, many types of applications for UAV require level cruising flight for a substantial distance, simultaneous with VTOL ability. VTOL ability is especially relevant when there is no dedicated runway, and accommodation of such runway is impossible due to the absence of clear landing space, for example in urban environment, marine vessels, drilling platforms, substantially uneven terrain etc.
- (5) In the art are known convertible aircrafts of several structural types that have been used in manned aviation, differing in the principles of flight in various modes. In particular, the so-called "convertiplane" are known, consisting of two rotor systems-lifting and cruising, each ones work in only one mode. The disadvantage of mentioned design is relatively high weight of propulsion unit, and degradation of aerodynamic characteristics caused by creation of additional drag of non-active rotors.
- (6) Likewise, there are known so-called "tiltrotors", providing variable flight modes by rotating (tilting) rotors from vertical position to horizontal configuration for the level cruising flight. Also there are known aerial vehicles providing variable flight modes by rotation of wing parts, on which rotors are mounted.
- (7) The disadvantage of the above mentioned structures of UAV is the instability of flight in transition modes, the structural complexity and high weight of the drivetrains providing rotation (tilting) rotors or wing parts. The mentioned structures are also poorly scaled, since with an increase in mass-dimensional characteristics, the pitch, roll and torque moments of rotors increase, which must be overcome structurally relative to the vertical flight plane.
- (8) Likewise, there are known so-called "tail sitter" VTOL, which have also been used in UAVs, however, as a rule, have a small relative carrying capacity and that are challenging to operate in circumstances of high cross winds in takeoff, landing and hovering modes.
- (9) Thus, the common disadvantages of the above-mentioned designs can be summarized as a) instability of the flight of the aircraft when making the transition from vertical to level cruising flight mode and back, as well as b) the difficulty of balancing the center of gravity of the aircraft, c) the complexity of control, and d) low reliability.
- (10) Examples for the above solutions are disclosed, inter alia in WO 2015/115913 A1, entitled "Multipurpose aircraft", having twin fuselage configuration with front and rear structural member panels, which are located between fuselages, wherein the front structural member panel includes a nacelle having compartment for storage and an engine. The front structural member panel described is part of the fixed wing assisting in balancing the lift of the aircraft.
- (11) Additionally, U.S. Pat. No. D822,579, design of aircraft, comprises a cabin with attached right and left wing consoles, right and left longitudinal beams attached by front outstanding struts to the

cabin and coupled together by airfoil element behind the cabin, motor with pushing rotor positioned on the back of the cabin, wherein each longitudinal beam includes row of four rotors for hover mode, vertical stabilizer positioned on the back of beam behind the cabin, landing gears. The disadvantage of aircraft shown, is that the one include two groups of rotors-one pushing rotor, positioned on the back of the cabin, to provide level cruising flight of the aircraft and two rows of rotors positioned on right and left beams, to provide hovering mode of the aircraft, thus increasing the weight of the aircraft and reducing energy efficiency and range.

(12) Likewise, U.S. Pat. No. 9,296,478 B2 entitled "Aircraft having at least two aircraft fuselages and a structural member with arrangement with at least two wing sections that are not connected to each other", includes a structural member arrangement with at least two non-connected wing sections, and a second wing, wherein connecting regions of the structural member arrangement and of the second wing are arranged so as to be offset relative to each other at least on an X-Z-plane on an aircraft-fixed coordinate system. The aircraft described in U.S. Pat. No. 9,296,478 B2 is a full-scaled passenger airplane providing level cruising flight, without vertical takeoff and landing capabilities.

(13) These and other shortcomings of the existing technology are sought to be resolved herein.

SUMMARY

(14) Disclosed, in various exemplary implementations, are hybrid unmanned aerial vehicles (UAVs), having vertical take-off and landing (VTOL) capabilities. Specifically, provided are exemplary implementations of hybrid fixed and rotating wings' UAV having VTOL capabilities with increased stability.

(15) In an exemplary implementation, provided herein is an unmanned aerial vehicle (UAV) system configured for Vertical Take-Off and Landing (VTOL), comprising an aircraft having: a pair of elongated arcuate drivetrain members, each having a basal end and an apical end and each defining a basal, mid, and apical inflection points; a fuselage; a structural member defining a longitudinal axis, having an upper surface and a basal surface, with a pair of lateral ends extending laterally from the fuselage and coupled to each of the elongated arcuate drivetrain members at each lateral end; a pair of second wings, operably coupled to, and extending laterally from each elongated arcuate drivetrain members, each second wing operably coupled to the structural member; a rear horizontal inverted airfoil, having apical surface and a basal surface spanning the gap between the pair of elongated arcuate drivetrain members, with lateral ends coupled to the pair of elongated arcuate drivetrain members at the apical inflection point; and optionally, a stabilizing cross bar having a pair of lateral ends coupled to the corresponding elongated arcuate drivetrain member at the basal inflection point, wherein each elongated drivetrain member further comprises a first VTOL rotor extending apically from the basal inflection point and a second VTOL rotor extending basally from the elongated arcuate drivetrain member between the mid inflection point and the apical inflection point.

(16) In another exemplary implementation, provided herein is an autonomous VTOL UAV comprising: a first autopilot module transceiving UAV rotor control signals; a second autopilot module receiving fixed wing control signals; an integrator module; and an on-board central processing module (CPM) in communication with the first autopilot module, the second autopilot module, and the integrator module, the CPM comprising at least one processor, and being in further communication with a non-transitory memory device storing thereon a set of executable instructions, configured, when executed to cause the at least one processor to automatically: transceive the rotor control signals; receive the fixed wing control signals; using the integrator module, calculate control signals to be applied to the VTOL UAV; and apply said calculated control signals to said VTOL UAV rotor controls.

(17) These and other features of the systems, methods and programs of the hybrid fixed and rotating wings UAV having VTOL capabilities, will become apparent from the following detailed

description when read in conjunction with the figures and examples, which are exemplary, not limiting.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) For a better understanding of systems, methods and programs of hybrid fixed and rotating wings UAVs having VTOL capabilities, with regard to the exemplary implementations thereof, reference is made to the accompanying examples and figures, in which:
- (2) FIG. 1A, illustrates a top perspective view of an exemplary implementation of the UAV with VTOL capabilities, with FIG. 1B, illustrating a bottom perspective view thereof;
- (3) FIG. 2, illustrates one of a pair of the drivetrain members;
- (4) FIG. 3A illustrating a top perspective view of the structural member element, with FIG. 3B, illustrating an enlarged portion;
- (5) FIG. 4A, illustrating a top perspective view the fuselage element of the UAV, with FIG. 4B, illustrating a bottom perspective view thereof, and FIG. 4C, illustrating the gondola portion of the fuselage, without the covering member;
- (6) FIG. 5A, illustrating the inverted air foil of the UAV, with FIG. 5B, illustrating a Y-Z cross section taken along line C-C of FIG. 5A;
- (7) FIG. 6A, illustrating side view of UAV in level cruising flight mode position with FIG. 6B, illustrating the UAV hovering at vertical landing and takeoff; and
- (8) FIG. 7, illustrates a top perspective view of another exemplary implementation of the UAV with VTOL capabilities, without the stabilizing cross-bar and the autopilot radiator illustrated in FIG. 1A.

DETAILED DESCRIPTION

- (9) Provided herein are exemplary implementations of systems, methods and programs of hybrid fixed and rotating wings UAV having VTOL capabilities. In certain exemplary implementations, the disclosed hybrid fixed and rotating wings UAV having VTOL capabilities, provides improved controllability, having seamless transitioning between hover and level cruising flight modes. The seamless transitioning between hover and level cruising flight modes is achieved, for example, by rotors mounted with fixed angle deflection in two arrays, wherein one array of front rotors are mounted with upward (apical) direction, and next array of rear rotors are mounted with downward (basal) direction, and the center of gravity of the UAV is located at the intersection of the diagonals of the rotor axes. The location of the center of gravity of the UAV and the values of the angles of rotor's deflection are determined by the equation provided herein.
- (10) As illustrated in an exemplary implementation in FIGS. 1A, 1B, Front wing arrangement with first rotors array and rear horizontal stabilizer (in other words, the inverted air foil) with second rotors (rotors) array affects two reacting thrust components, such that air lift area of front wing arrangement is larger than the lift generated by the rear horizontal stabilizer air lift area, creating a self-stabilizing aerodynamic system. In the configuration illustrated, the control of the UAV increases the nose pitch while decreasing downthrust by employing a functional elevator with a dedicated drive (rear rotor array). In the context of the disclosure, the term “rotor” is used to include rotors, propellers and any other suitable rotating blade or blade-type structure that imparts a force to a vehicle via interaction with the surrounding medium, be it air or fluid. Multi rotor UAV **10** can include multiple subsystems, for example an avionics subsystem, a genset subsystem, one or more of electronic speed controllers (ESCs) drive motors that drive one or more rotors (e.g., propellers). In some exemplary implementations, a drive motor is “coupleable” to a rotor/propeller. That is, the drive motor is adapted in a structure that is capable of being coupled to the rotor/propeller.

(11) Accordingly, provided is a hybrid UAV with VTOL capabilities, with two separated load bearing longitudinal elements, at least one front horizontal wing arrangement, rear horizontal stabilizer, multirotor propulsion unit, and fuselage for payload and other equipment. In certain exemplary implementations, the front wing arrangement comprises right and left separated wing parts without center wing section, which are mounted on outsides of the load bearing longitudinal drivetrain elements. In addition, the load bearing longitudinal drivetrain elements are optionally interconnected to each other by at least one transverse cross-bar stiffener.

(12) In the context of the disclosure, the term “transverse stiffener”, or “stabilizing cross bar” means a rod, bar or beam, or structure operable to provide for transverse stiffness of the front wing arrangement. The element is optionally included and its inclusion will depend on predetermined factors, such as payload weight, expected atmospheric conditions, and the like. As illustrated in FIG. 1B, the arcuate, flared V-shape cross bar is sized such that the apex of the V-shape stabilizer is located directly beneath the UAV's center of gravity, thus providing additional stability to the UAV. Likewise, in the context of the disclosure, the term “longitudinal element”, or “elongated arcuate drivetrain members” refer to load bearing element, operable for providing structural stiffness and for mounting equipment or for empennage. The term “drivetrain” means the mechanical, and electrical parts which interconnect the rotors mounted on the elongated arcuate drivetrain members to the power source.

(13) In the context of the disclosure, the term “operable” means the system and/or the device and/or the program, or a certain element or step is fully functional, sized, adapted and calibrated, comprises elements for, and meets applicable operability requirements to perform a recited function when activated, coupled, implemented, actuated, effected, realized, or when an executable program is executed by at least one processor associated with the system and/or the device. In relation to systems and circuits, the term “operable” means the system and/or the circuit is fully functional and calibrated, comprises logic for, having the hardware and firmware necessary, as well as the circuitry for, and meets applicable operability requirements to perform a recited function when executed by at least one processor.

(14) In another exemplary implementation, the multirotor propulsion unit comprises at least four rotors (interchangeable with rotors). The rotors can be located at least in two arrays, the first array ahead of the front wing arrangement and the second array forward from the horizontal stabilizer. The rotors are mounted on longitudinal drivetrain elements, with fixed deflection angle β_{iio} , $\beta_{sub.i20}$ and with a predetermined gap from the leading edge of the front wing arrangement and below the low pressure zone of the rear horizontal stabilizer. The rotational axes of first array of rotors are directed upward, and the rotational axes of second array of rotors are directed downward.

(15) Furthermore, the UAV further comprises rear horizontal stabilizer (or, rear horizontal inverted airfoil) comprised of spoiler and optionally, in certain exemplary implementations, elevator with drive. The spoiler being inverted and being operable to create a downward directed aerodynamic force (downthrust force). Moreover, the air gap between airfoils and rotors' tips sized and adapted to minimize distortion of the air flow created by the rotors and the airflow above and below the fixed wing assembly and the rear horizontal inverted airfoil.

(16) As further illustrated, e.g., in FIG. 6B, the UAV's center of gravity during vertical takeoff and landing is disposed directly beneath the intersection of diagonals drawn through the rotational axes of the rotors in each elongated arcuate drivetrain member, though not forming a pyramid, but rather on each side of the fuselage, thereby creating two intersecting components of thrust, acting to stabilize the UAV on VTOL mode.

(17) As illustrated in FIGS. 6A, 6B, the fixed deflection angle of the rotor's are operable to provide flight stability in both hovering and level cruising flight of the UAV. The stability, provided by the UAV systems disclosed, is determined using the following equation:

$$P_{1v} = P_{2v}$$

(18) $\left\{ \begin{array}{l} \cos = \times \cos \\ = \frac{L_2}{L_1} \end{array} \right.$ (EQU1) Wherein:—P.sub.1v, P.sub.2v—are vertical thrust

components of front **110, 110'** and rear **120, 120'** rotors; L1, L2, —moment arms laid between axes of vertical thrust components and center of gravity (COG); β .sub.110, β .sub.120—estimated values of angles of design deflection of rotors, producing normal rotor thrust P.sub.1n, P.sub.2n; and k—design factor

(19) As indicated, during hovering mode, stability is provided by convergence of thrust vectors of front **110, 110'** and rear **120, 120'** rotors in intersection point over the UAV, while the UAV's COG is disposed below the intersection point as shown in FIG. 6B.

(20) In certain exemplary implementation, the fuselage further comprises auto piloting system and a gondola, for example, for mounting imaging module equipment, or for storing another payload.

(21) It is contemplated, that the UAV disclosed, can be scaled up to be manned aerial vehicle, operable to carry personnel, passengers and crew, as well as payload.

(22) A more complete understanding of the components, and devices disclosed herein can be obtained by reference to the accompanying drawings. These figures (also referred to herein as "FIG.") are merely schematic representations based on convenience and the ease of demonstrating the present disclosure and are, therefore, not intended to indicate relative size and dimensions of the devices or components thereof, their relative size relationship and/or to define or limit the scope of the exemplary implementations. Although specific terms are used in the following description for the sake of clarity, these terms are intended to refer only to the particular structure of the exemplary implementations selected for illustration in the drawings, and are not intended to define or limit the scope of the disclosure relative to the rest of the specification. In the drawings and the following description below, it is to be understood that like numeric designations refer to components of like function. Likewise, cross sections are referred to on normal orthogonal coordinate system having XYZ axis, such that Y axis refers to front-to-back, X axis refers to side-to-side, and Z axis refers to up-and-down.

(23) Turning now to FIGS. 1 A-5B, illustrating unmanned aerial vehicle (UAV) system configured for Vertical Take-Off and Landing (VTOL), comprising: aircraft **10** comprising: pair of elongated arcuate drivetrain members **100, 100'**, each having basal end **101, 101'** with basal pad **1010, 1010'** optionally coupled, and apical end **102, 102'** and each defining basal **1001** (see e.g., FIG. 2), mid **1002**, and apical **1003** inflection points; fuselage **200**; structural member **300** defining longitudinal axis A/, (See e.g., FIGS. 3A, 3B) having upper surface **3003** and basal surface **3004**, with pair of lateral end caps **3001, 3002** coupled to lateral sections **3006, 3007** extending laterally from fuselage **200** and coupled to each of elongated arcuate drivetrain members **100, 100'** at each lateral end-cap **3001, 3002** respectively. Also illustrated, are pair of second wings **400, 400'**, operably coupled to, and extending laterally from each elongated arcuate drivetrain members **100, 100'**, each second wing **400, 400'** operably coupled to structural member's **300** via, for example at least one tube **319A, 329A** operable to couple the components through apertures **1009A, 1009A'**, with opening **1008 (1008')** (See e.g., FIG. 2) defined in each elongated arcuate drivetrain member **100, 100'** being used for example, for wirings.

(24) UAV **10** further comprises rear horizontal inverted airfoil **500**, having leading edge **5003** and trailing edge **5004**, with apical surface **5001** and basal surface **5002** spanning the gap between the pair of elongated arcuate drivetrain members **100, 100'**, with lateral ends **5005, 5005'** coupled to pair of elongated arcuate drivetrain members **100, 100'** at or about apical inflection point **1003** (see e.g., mounting pad **1006**, FIG. 1B). Also shown is optional, stabilizing cross bar **130** having pair of lateral ends **1300, 1300'** (see e.g., FIGS. 1A and 1B) coupled to corresponding elongated arcuate drivetrain member **100, 100'** at or about basal inflection point **1001**, wherein each elongated

drivetrain member **100, 100'** further comprises first VTOL rotor **110, 110'** extending apically from basal inflection point **1001, 100G** and second VTOL rotor **120, 120'** extending basally from elongated arcuate drivetrain member **100, 100'** between mid-inflection point **1002, 1002'** and apical inflection point **1003, 1003'**. As illustrated in FIG. 7, under certain circumstances, stabilizing cross bar **130** is not incorporated into the system, and therefore reduce the weight (thus increasing the range) of the UAV.

(25) Moreover, and as illustrated in FIG. 2, each elongated arcuate drivetrain member **100, 100'** further comprise dorsal vertical stabilizer (vertical air foil) **105, 105'** extending dorsally from mid inflection **1002, 1002'** point to apical inflection point **1003, 1003'**, and ventral horizontal stabilizer **106, 106'** disposed between apical infection point **1003, 1003'** and apical end **102, 102'**. Moreover, and as illustrated in FIG. 2, each elongated arcuate drivetrain member **100, 100'** further comprises support member **103, 103'** having basal end **1031, 1031'** (for example, a pad) extending basally from mid inflection point **1002, 1002'**, operable to form level base plane **101, 101', 103', 103** at level cruising flight and a level plane **103, 102, 102', 103'** upon VTOL (see e.g., FIG. 6B). The elongated arcuate drivetrain members **100, 100'** can be monolithic light weight hollow body structure, made of, for example, polypropylene, ABS plastic, or a carbon fiber/Kevlar composite.

(26) As further illustrated, rotors **110, 110' 120, 120'** are disposed at predetermined deflection angles bpo, bho such that at takeoff, while plane **103, 102, 102', 103'**, the projected diagonal formed from the rotational axis of each rotor **110, 120, 110', 120'**, intersects normal to the UAV center of gravity. In an exemplary implementation, bho can be, for example between about 45° and about 47°, while bho can be between about 41° and about 43°, such that first VTOL rotor **110, 110'** and second VTOL rotor **120, 120'**, each extend from elongated arcuate drivetrain member **100, 100'** at predetermined deflection angle bpo, b.sub.120 off vertical, and wherein second VTOL rotor **120, 120'** extends from elongated arcuate drivetrain member at larger deflection angle fim than first VTOL rotors '110, 110' deflection angle bho,. Furthermore, depending on various parameters, such as VTOL altitude, payload weight, atmospheric conditions and the like, the rotor blades can be replaced from, for example in one exemplary implementation 25.4 cm. diameter with a step of about 11.4 cm., to a rotor having a blade diameter of about 28 cm. with a step of about 11.4 cm. As further illustrated in FIG. 2, each rotor **110, 110', 120, 120'** are mounted on corresponding mounting pads **1004, 1005, 1004', 1005'**.

(27) Turning now to FIGS. 2-3B, illustrating (aerodynamically contoured) structural member **300**, having lateral end caps **3001, 3002** with leading edge **3008** and trailing edge **3009** further defining central region **310** and pair of lateral sections **3006, 3007** tapering away from central region **310**, while upper surface **3100** of central region **310** further defines front portion **3101** and rear portion **3102**, rear portion **3102** further defining opening **3103**, operable to accommodate equipment and communicate with fuselage covering **210** (see e.g., FIG. 4A). As illustrated in FIG. 3A front portion **3101** and rear portion **3102**, are raised over upper surface **3003** of structural member **300**, forming a predetermined topology. Also illustrated in FIG. 3B, is aperture **3105**, operable to provide access to structural member **300**, for coupling of various elements, such as pressure sensors and the like (**311**, see e.g., FIG. 1A), for example, downward looking LIDAR **350** (see e.g., FIG. 1B).

(28) As further illustrated in FIGS. 2, 3 A, and 3B, each end-cap **3001, 3002** is operable to accommodate tubes **319A, 319B, 329 A, 329B** sized, adapted and configured, in certain exemplary implementation, to couple second wing section **400, 400'**, through apertures **1009A, 1009B, 1009A', 1009B'**. Likewise, bores **1011A, 1011B, 1011A', 1011B'** are configured to accommodate retainers **321A, 321B, 322A, 322B** (e.g., screws, detents, thus coupling structural member **300** to each elongated drivetrain member **100, 100'**. As illustrated further, each end-cap **3001, 3002** further defines opening **3021, 3022**, configured to allow passage between second wings **400, 400'** and fuselage **200**, doe example, wiring.

(29) Turning now to FIGS. 4A-4C, illustrating fuselage **200** comprising: basally open covering member **210**, and gondola **220**, together forming a nacelle. As illustrated, basally open covering

member **210** having anterior end **2101** and posterior end **2102** comprises nose section **211**, canopy section **212** and backing section **213** together defining lip **2105**, lip **2105** forms an opening sized and adapted to accommodate central region **310** topology formed of front portion **3101** and rear portion **3102** of structural member **300** between covering member **210** and gondola **220**, while gondola **220** comprises: posteriorly open aft portion **2202**, forming compartment **2205** operable to accommodate power unit **260**; with apically open fore portion **2206**, forming payload chamber, aft portion **2205** separated from fore portion **2206** by bulkhead **2207** (not shown).

(30) Also shown in FIGS. **4B**, are optional opening **221**, which can be used to enable imaging module **700** to observe the ground. Also illustrated in FIG. **4B**, are means **222** for coupling gondola **220** having basal surface **2204**, to transverse stabilizing bar **130** (see e.g., FIG. **1A**). The coupling means can be any suitable means, such as screws, rods, detents, zip ties, and the like. As further illustrated in FIG. **4C**, gondola **220**, having anterior end **2201**, posterior end **2202**, basal surface **2204**, and apical surface **2203** with side walls **2208**, is configured in certain exemplary implementation to have upper deck **228** and lower deck **229**, separated by step **2285**. Apically open fore section **2206**, further shows payload **240** (here a camera clamp, but could be other payload type), with power unit releasing lever **226** and modem **227** operable to communicate with the ground control station in certain exemplary implementations. Lower deck **229**, further shows autopilot module **230** with cooling radiator **250** coupled to lower deck **229**, as do coupling means **2291i**, operable to couple gondola **220** to structural member **300** central portion **310**. Charging port **2601**, as well as leads **2600** to power unit **260** are also shown as is closure clasp **223**. While autopilot **230** provides autonomous flight capabilities. In certain other exemplary implementations, the UAV can be remote controlled and radiator **250** will become unnecessary as further illustrated in FIG. **7**.

(31) Turning now to FIG. **5 A**, **5B**, illustrating rear horizontal inverted airfoil **500** having negative angle of attack, of between about -1.0° and -5.0° , for example, between about -2° and about -3° . As illustrated, in certain exemplary implementations, the ratio between the span (S_{soo}) and chord length (Ch_{soo}) is operable to improve the stability of the UAV, and may depend on, for example, design parameter such as the length of rotor blade (preventing flow distortion), maintain the optimal air gap, as well as the fixed wing assembly comprised of structural member **300** and second wings **400**, **400'**, and their span.

(32) In an exemplary implementation, the UAV disclosed, is used as an autonomous UAV having VTOL capabilities. Accordingly, provided herein is an autonomous VTOL UAV **10** comprising: a first autopilot module transceiving (in other words, operable to transmit and receive) UAV rotor control signals; a second autopilot module receiving fixed wing control signals; an integrator module; and an on-board central processing module (CPM) in communication with the first autopilot module, the second autopilot module, and the integrator module, the CPM comprising at least one processor, and being in further communication with a non-transitory memory device storing thereon a set of executable instructions, configured, when executed to cause the at least one processor to automatically: transceive the rotor control signals; receive the fixed wing control signals; using the integrator module, calculate control signals to be applied to the VTOL UAV; and apply said calculated control signals to said VTOL UAV rotor controls. Consequently, VTOL UAV comprises: pair of elongated arcuate drivetrain members **100**, **100'**, each having basal end **101**, **101'** and apical end **102**, **102'** and each defining basal **1001**, **1001'**, mid **1002**, **1002'**, and apical **1003**, **1003'** inflection points; fuselage **200**; structural member **300** defining longitudinal axis XL, having upper surface **3006** and basal surface **3007**, with pair of lateral ends **3001**, **3002** extending laterally from fuselage **200** and coupled to each of elongated arcuate drivetrain members **100**, **100'** at each lateral end **3001**, **3002** respectively. Pair of second wings **400**, **400'** are operably coupled to, and extending laterally from each elongated arcuate drivetrain members **100**, **100'**, each second wing **400**, **400'** operably coupled to structural member 's **300** longitudinal axis A/, (e.g., through a pair of tubes operable to be accommodated in apertures **1009 A**, **1009B** and **1009A'**, **1009B'** (see e.g., FIG.

2)). Rear horizontal inverted airfoil **500**, having apical surface **5001** and basal surface **5002** spanning gap between pair of elongated arcuate drivetrain members **100**, **100'**, with lateral ends **5005**, **5005'** coupled to pair of elongated arcuate drivetrain members **100**, **100'** at apical inflection point **1003** mounting **1006**, wherein structural member **300**, pair of second wings **400**, **400'** together (with tubes **321A**, **321B**, **321A'**, **321B'**, **322A**, **322B**, **322A'**, **322B'** [not shown]) coupling each of second wings **400**, **400'** to structural member **300** through apertures **1009A**, **1009B** and **1009A'**, **1009B'** defined in elongated arcuate members **100**, **100'**, for example passing through opening **1008** [see e.g., FIG. 2,] defined in elongated arcuate members **100**, **100'**) form fixed wing assembly such that fixed wing assembly, and rear horizontal inverted airfoil **500** is operable to transmit plurality of control signals. In certain exemplary implementations, the plurality of fixed wing assembly control signals comprise, for example, at least one of: pressure, temperature, wind speed above the fixed wing, pitch angle, roll angle, and yaw angle, some of which can be transmitted using antennae **141**, **142**. Depending on the task, UAV **10** can further optionally comprise, stabilizing cross bar **130** having pair of lateral

(33) ends **1300**, **1300'** coupled to corresponding elongated arcuate drivetrain member **100**, **100'** at basal inflection point **1001** (see e.g., FIG. 2, **1007**), wherein each elongated drivetrain member **100**, **100'** further comprises first VTOL rotor **110**, **110'** extending apically from basal inflection point **1001**, **100G** and second VTOL rotor **120**, **120'** extending basally from elongated arcuate drivetrain member **100**, **100'** between mid inflection point **1002**, **1002'** and apical inflection point **1003**, **1003'**, first **110**, **110'** and second **120**, **120'** rotor operable to transceive plurality of rotor control signals, the plurality of rotor control signal comprise at least one of: pressure, temperature, ground speed, revolution per minute (RPM), and altitude.

(34) Optional stabilizing bar **130**, is depicted having antennae **141**, **142** coupled thereto, operable to transmit signals as needed to autopilot (see e.g., **230**, FIG. 4C), as well as

(35) The UAV systems disclosed herein can be computerized systems further comprising a central processing module (CPM); a display module; and a user interface module. The Display modules, which can include display elements, which may include any type of element which acts as a display. A typical example is a Liquid Crystal Display (LCD). LCD for example, includes a transparent electrode plate arranged on each side of a liquid crystal. There are however, many other forms of displays, for example OLED displays and Bi-stable displays. New display technologies are also being developed constantly. Therefore, the term display should be interpreted widely and should not be associated with a single display technology. Also, the display module may be mounted on a printed circuit board (PCB) of an electronic device, arranged within a protective housing and the display module is protected from damage by a glass or plastic plate arranged over the display element and attached to the housing.

(36) The term “communicate” (and its derivatives e.g., a first component “communicates with” or “is in communication with” a second component) and grammatical variations thereof are used to indicate a structural, functional, mechanical, electrical, optical, or fluidic relationship, or any combination thereof, between two or more components or elements. As such, the fact that one component is said to communicate with a second component is not intended to exclude the possibility that additional components can be present between, and/or operatively associated or engaged with, the first and second components. Furthermore, the term “electronic communication” means that one or more components of the multi-mode optoelectronic observation and sighting system with cross-platform integration capability described herein are in wired or wireless communication or internet communication so that electronic signals and information can be exchanged between the components.

(37) Likewise, the term “module” is understood to encompass a tangible entity, be that an entity that is physically constructed, specifically configured (e.g., hardwired), or temporarily (e.g., transitorily) configured (e.g., programmed) to operate in a specified manner or to perform part or all of any operation described herein. Considering examples in which modules are temporarily

configured, each of the modules need not be instantiated at any one moment in time. For example, where the modules comprise a general-purpose hardware processor configured using software, the general-purpose hardware processor may be configured as respective different modules at different times. Software may accordingly configure a hardware processor, for example, to constitute a particular module at one instance of time and to constitute a different module at a different instance of time. In an embodiment, an electronic control unit of the systems disclosed and claimed, is the electronic control module (ECM).

(38) The term “computer-readable medium” as used herein, in addition to having its ordinary meaning, refers to any medium that participates in providing instructions to a processor for execution. Such a medium may take many forms, including but not limited to, non-volatile media and volatile media. Non-transitory media can be, for example, optical or magnetic disks, such as a storage device. Volatile media includes dynamic memory, such as main memory.

(39) Memory device as used in the methods, programs and systems described herein can be any of various types of memory devices or storage devices. The term “memory device” is intended to encompass an installation medium, e.g., a CD-ROM, floppy disks, or tape device; a computer system memory or random access memory such as DRAM, DDR RAM, SRAM, EDO RAM, Rambus RAM, etc.; or a non-volatile memory such as a magnetic media, e.g., a hard drive, optical storage, or ROM, EPROM, FLASH, etc. The memory device may comprise other types of memory as well, or combinations thereof. In addition, the memory medium may be located in a first computer in which the programs are executed (e.g., the UAV on-board CPM), and/or may be located in a second different computer [or micro controller, e.g., the ground control unit] which connects to the first computer over a network, such as cellular network, satellite, wireless network or their combination (Mesh networks). In the latter instance, the second computer may further provide program instructions to the first computer for execution. The term “memory device” can also include two or more memory devices which may reside in different locations, e.g., in different computers that are connected over a network.

(40) The term “comprising” and its derivatives, as used herein, are intended to be open ended terms that specify the presence of the stated features, elements, components, groups, integers, and/or steps, but do not exclude the presence of other unstated features, elements, components, groups, integers and/or steps. The foregoing also applies to words having similar meanings such as the terms, “including”, “having” and their derivatives.

(41) The terms “a”, “an” and “the” herein do not denote a limitation of quantity, and are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The suffix “(s)” as used herein is intended to include both the singular and the plural of the term that it modifies, thereby including one or more of that term (e.g., the stack(s) includes one or more stack). Reference throughout the specification to “one exemplary implementation”, “another exemplary implementation”, “an exemplary implementation”, and so forth, when present, means that a particular element (e.g., feature, structure, and/or characteristic) described in connection with the exemplary implementation is included in at least one exemplary implementation described herein, and may or may not be present in other exemplary implementations. In addition, it is to be understood that the described elements may be combined in any suitable manner in the various exemplary implementations.

(42) Unless specifically stated otherwise, as apparent from the discussions, it is appreciated that throughout the specification discussions utilizing terms such as “processing,” “loading,” “in communication,” “detecting,” “calculating,” “determining”, “analyzing,” “applying” or the like, refer to the action and/or processes of a computer or computing system, or similar electronic computing device, that manipulate and/or transform data represented as and to physical manifestation.

(43) For the purposes of the present invention, directional or positional terms such as “top”, “bottom”, “upper,” “lower,” “side,” “front,” “frontal,” “forward,” “rear,” “rearward,” “back,”

“trailing,” “above,” “below,” “left,” “right,” “horizontal,” “vertical,” “upward,” “downward,” “outer,” “inner,” “exterior,” “interior,” “intermediate,” etc., are merely used for convenience in describing the various embodiments of the present invention. For example, the orientation of the embodiments shown in FIGS. 1-6 may be reversed or flipped over, rotated by 90 degrees in any direction, etc.

(44) Accordingly and in an exemplary implementation, provided herein is an unmanned aerial vehicle (UAV) system configured for Vertical Take-Off and Landing (VTOL), comprising: an aircraft comprising: a pair of elongated arcuate drivetrain members, each having a basal end and an apical end and each defining a basal, mid, and apical inflection points; a fuselage; a structural member defining a longitudinal axis, having an upper surface and a basal surface, with a pair of lateral ends extending laterally from the fuselage and coupled to each of the elongated arcuate drivetrain members at each lateral end; a pair of second wings, operably coupled to, and extending laterally from each elongated arcuate drivetrain members; a rear horizontal inverted airfoil, having apical surface and a basal surface spanning the gap between the pair of elongated arcuate drivetrain members, with lateral ends coupled to the pair of elongated arcuate drivetrain members at the apical inflection point; and optionally, a stabilizing cross bar having a pair of lateral ends coupled to the corresponding elongated arcuate drivetrain member at the basal inflection point, wherein each elongated drivetrain member further comprises a first VTOL rotor extending apically from the basal inflection point and a second VTOL rotor extending basally from the elongated arcuate drivetrain member between the mid inflection point and the apical inflection point, wherein (i) each elongated arcuate drivetrain member further comprises a dorsal vertical stabilizer extending from the mid inflection point to the apical inflection point, and a ventral horizontal stabilizer disposed between the apical inflection point and the apical end, (ii) further comprising a support member extending basally from the mid inflection, operable to form a level base plane, wherein (iii) the first VTOL rotor and the second VTOL rotor, each extend from the elongated arcuate drivetrain member at a predetermined deflection angle off vertical, and wherein the second VTOL rotor extends from the elongated arcuate drivetrain member at a larger deflection angle than the first VTOL rotor's deflection angle, wherein (iv) the structural member further defining a central region and a pair of lateral sections tapering away (in other words, becoming narrower the further the distance from) from the central region, wherein (v) the upper surface of the central region further defines a front portion and a rear portion, the rear portion further defining an opening, wherein (vi) the fuselage comprises: a basally open covering member; and a gondola (both forming a nacelle), (vii) the basally open covering member comprises a nose section, a canopy section and a backing section together defining a lip and wherein the lip forms an opening sized and adapted to accommodate the structural member between the basally open covering member and the gondola, (viii) the gondola comprises: a posteriorly open aft portion, forming a compartment operable to accommodate a power unit; an apically open fore portion, forming a payload chamber, the aft portion separated from the fore portion by a bulkhead, and wherein (ix) the rear horizontal inverted airfoil has a negative angle of attack, of between -1° and -5° .

(45) In another exemplary implementation, provided herein is an autonomous VTOL UAV comprising: a first autopilot module transceiving (in other words, operable to transmit and receive) UAV rotor control signals; a second autopilot module receiving fixed wing control signals; an integrator module; and an on-board central processing module (CPM) in communication with the first autopilot module, the second autopilot module, and the integrator module, the CPM comprising at least one processor, and being in further communication with a non-transitory memory device storing thereon a set of executable instructions, configured, when executed to cause the at least one processor to automatically: transceive the rotor control signals; receive the fixed wing control signals; using the integrator module, calculate control signals to be applied to the VTOL UAV; and apply said calculated control signals to said VTOL UAV rotor controls, wherein (x) the VTOL UAV comprises: a pair of elongated arcuate drivetrain members, each having a basal

end and an apical end and each defining a basal, mid, and apical inflection points; a fuselage; a structural member defining a longitudinal axis, having an upper surface and a basal surface, with a pair of lateral ends extending laterally from the fuselage and coupled to each of the elongated arcuate drivetrain members at each lateral end; a pair of second wings, operably coupled to, and extending laterally from each elongated arcuate drivetrain members, each second wing further operably coupled to the structural member; a rear horizontal inverted airfoil, having apical surface and a basal surface spanning the gap between the pair of elongated arcuate drivetrain members, with lateral ends coupled to the pair of elongated arcuate drivetrain members at the apical inflection point, wherein the structural member, pair of second wings, and the rear horizontal inverted airfoil form the fixed wing, wherein the fixed wing is operable to transmit a plurality of control signals; and optionally, a stabilizing cross bar having a pair of lateral ends coupled to the corresponding elongated arcuate drivetrain member at the basal inflection point, wherein each elongated drivetrain member further comprises a first VTOL rotor extending apically from the basal inflection point and a second VTOL rotor extending basally from the elongated arcuate drivetrain member between the mid inflection point and the apical inflection point, the first and second rotor operable to transceive a plurality of rotor control signals, wherein (xi) each elongated arcuate drivetrain member further (optionally) comprises a dorsal vertical stabilizer extending the mid inflection point to the apical inflection point, and (optionally) a ventral horizontal stabilizer disposed between the apical inflection point and the apical end, (xii) as well as a support member extending basally from the mid inflection, operable to form a level base (tail) plane, wherein (xiii) the first VTOL rotor and the second VTOL rotor, each extend from the elongated arcuate drivetrain member at a predetermined deflection angle off vertical, and wherein the second VTOL rotor extends from the elongated arcuate drivetrain member at a larger angle than the first VTOL rotor and wherein the plurality of rotor control signal comprise at least one of: pressure, temperature, ground speed, revolution per minute (RPM), and altitude, wherein (xiv) the structural member further defining a central region and a pair of lateral regions tapering away from the central region; the upper surface of the central region further defines a front portion and a rear portion, the rear portion further defining an opening; the plurality of fixed wing control signals comprise at least one of: pressure, temperature, wind speed above the fixed wing, pitch angle, roll angle, and yaw angle, wherein (xv) the fuselage comprises: a basally open covering member; and a gondola, (xvi) the basally open covering member comprises a nose section, a canopy section and a backing section together defining a lip and wherein the lip forms an opening sized and adapted to accommodate the structural member between the basally open covering member and the gondola, (xvii) the gondola comprises: a posteriorly open aft portion, forming a compartment operable to accommodate a power unit, the compartment having an upper deck surface; an autopilot element, comprising the first autopilot module, the second autopilot module, the integrator module, a communication module, and a navigation module; and an apically open fore portion, forming a payload chamber, the aft portion separated from the fore portion by a bulkhead, and wherein (xviii) the rear horizontal inverted airfoil has a negative angle of attack, of between -1° and -5°

(46) While the invention has been described in detail and with reference to specific exemplary implementations thereof, it will be apparent to one of ordinary skill in the art that various changes and modifications can be made therein without departing from the spirit and scope thereof. Accordingly, it is intended that the present disclosure covers the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

Claims

1. An unmanned aerial vehicle (UAV) system configured for Vertical Take-Off and Landing (VTOL), comprising: an aircraft comprising: a) a pair of elongated arcuate drivetrain members, each having a basal end and an apical end and each defining a basal, mid, and apical inflection

points, wherein each elongated arcuate drivetrain member further comprises a first VTOL rotor extending apically from the basal inflection point and a second VTOL rotor extending basally from the elongated arcuate drivetrain member between the mid inflection point and the apical inflection point; b) a fuselage; c) a structural member defining a longitudinal axis, having an upper surface and a basal surface, with a pair of lateral ends extending laterally from the fuselage and coupled to each of the elongated arcuate drivetrain members at each lateral end; d) a pair of fixed wings, operably coupled to, and extending laterally from each elongated arcuate drivetrain members; e) a rear horizontal inverted airfoil, having apical surface and a basal surface spanning a gap between the pair of elongated arcuate drivetrain members, with lateral ends coupled to the pair of elongated arcuate drivetrain members at the apical inflection point; and f) a stabilizing cross bar having a pair of lateral ends coupled to the corresponding elongated arcuate drivetrain member at the basal inflection point.

2. The UAV system of claim 1, wherein each elongated arcuate drivetrain member further comprises a dorsal vertical stabilizer extending from the mid inflection point to the apical inflection point, and a ventral horizontal stabilizer disposed between the apical inflection point and the apical end.

3. The UAV of claim 2, wherein each elongated arcuate drivetrain member further comprises a support member extending basally from the mid inflection, operable to form a level base plane.

4. The UAV of claim 3, wherein the first VTOL rotor and the second VTOL rotor, each extend from the elongated arcuate drivetrain member at a predetermined deflection angle off vertical, and wherein the second VTOL rotor extends from the elongated arcuate drivetrain member at a larger deflection angle than the first VTOL rotor's deflection angle.

5. The UAV of claim 4, further comprising: a) a first autopilot module transceiving UAV rotor control signals; b) a second autopilot module receiving fixed wing control signals; c) an integrator module; and d) an on-board central processing module (CPM) in communication with the first autopilot module, the second autopilot module, and the integrator module, the on-board CPM comprising at least one processor, and being in further communication with a non-transitory memory device storing thereon a set of executable instructions, configured, when executed to cause the at least one processor to automatically: i. transceive the rotor control signals; ii. receive the fixed wing control signals; iii. using the integrator module, calculate control signals to be applied to the VTOL UAV; and iv. apply said calculated control signals to said VTOL UAV rotor controls.

6. The UAV system of claim 5, wherein, the plurality of fixed wing control signals comprise at least one of: pressure, temperature, wind speed above the fixed wing, pitch angle, roll angle, and yaw angle.

7. The UAV system of claim 5, wherein the plurality of rotor control signals comprise at least one of: pressure, temperature, ground speed, revolution per minute (RPM), and altitude.

8. The UAV of claim 1, wherein the structural member further defining a central region and a pair of lateral sections tapering away from the central region.

9. The UAV of claim 8, wherein the upper surface of the central region further defines a front portion and a rear portion, the rear portion further defining an opening.

10. The UAV of claim 1, wherein the fuselage comprises: a. a basally open covering member; and b. a gondola.

11. The UAV of claim 10, wherein the basally open covering member comprises a nose section, a canopy section and a backing section together defining a lip and wherein the lip forms an opening sized and adapted to accommodate the structural member between the basally open covering member and the gondola.

12. The UAV of claim 11, wherein the gondola comprises: a) a posteriorly open aft portion, forming a compartment operable to accommodate a power unit; and b) an apically open fore portion, forming a payload chamber, the aft portion separated from the fore portion by a bulkhead.

13. The UAV of claim 1, wherein the rear horizontal inverted airfoil has a negative angle of attack,

of between -1° and -5° .

14. The UAV of claim 1, wherein the UAV's center of gravity (COG) during vertical takeoff and landing is disposed directly beneath an intersection of projected diagonals drawn through the rotational axes of the first VTOL rotor and the second VTOL rotor in each elongated arcuate drivetrain member.
